

POGRANICHNOYE, Russian Federation

WMO: 320760

Lat: 50.400N

Lon: 143.767E

Elev: 8

StdP: 101.23

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.0	(c) -25.8	(d) -30.9	(e) 0.2	(f) -27.1	(g) -28.9	(h) 0.3	(i) -24.8	(j) 10.9	(k) -5.7	(l) 9.3	(m) -5.2	(n) 2.1	(o) 280	(p) 0.737

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.5	(c) 22.3	(d) 16.4	(e) 19.5	(f) 15.3	(g) 17.4	(h) 14.3	(i) 17.4	(j) 20.9	(k) 16.0	(l) 18.7	(m) 14.8	(n) 16.9	(o) 3.2	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	11.4	18.6	14.7	10.5	17.2	13.6	9.7	15.9	49.3	21.2	45.1	18.8	41.5	17.0	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.0	(b) 6.4	(c) 5.4	(e) -31.9	(f) 29.1	(g) 2.2	(h) 2.8	(i) -33.5	(j) 31.1	(k) -34.8	(l) 32.8	(m) -36.0	(n) 34.3	(o) -37.7	(p) 36.4		
(a) 8.0	(b) 6.4	(c) 5.4	(e) -31.9	(f) 21.0	(g) 2.3	(h) 1.6	(i) -33.5	(j) 22.1	(k) -34.9	(l) 23.1	(m) -36.2	(n) 24.0	(o) -37.8	(p) 25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.7	-14.6	-14.9	-9.2	-1.8	2.9	7.4	10.5	12.5	10.6	4.2	-4.8	-12.3
	DBStd	10.65	5.65	5.31	4.84	3.54	3.37	3.53	2.91	3.00	3.14	3.24	5.27	5.48
	HDD10.0	4111	764	696	596	355	224	95	26	7	29	182	446	691
	HDD18.3	6963	1022	929	854	605	479	329	243	184	233	439	696	949
	CDD10.0	194	0	0	0	0	3	17	43	83	47	2	0	0
	CDD18.3	5	0	0	0	0	0	1	1	3	0	0	0	0
	CDH23.3	60	0	0	0	0	1	16	11	27	6	0	0	0
	CDH26.7	13	0	0	0	0	0	5	3	4	1	0	0	0
Wind	WSAvg	2.3	2.7	2.5	2.4	2.3	2.2	2.0	1.8	1.8	2.0	2.5	2.5	2.7
Precipitation	PrecAvg	846	52	34	53	77	71	61	57	95	99	99	86	63
	PrecMax	1200	115	89	128	236	178	135	132	259	316	287	255	212
	PrecMin	514	7	8	4	5	11	10	3	23	28	11	15	2
	PrecStd	176	32	20	36	52	44	34	40	50	66	64	58	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.7	-1.0	3.8	12.7	19.6	25.4	24.3	26.8	23.7	17.9	8.2	1.1
		MCWB	-1.2	-2.1	0.2	6.2	11.3	15.7	18.0	18.9	15.6	11.1	4.7	0.5
	2%	DB	-1.5	-2.4	0.6	7.8	14.6	18.5	19.2	22.5	20.7	14.3	5.4	-0.3
		MCWB	-2.0	-3.3	-1.3	3.7	8.9	13.4	16.3	17.4	15.0	9.6	3.5	-0.6
	5%	DB	-2.6	-3.8	-0.7	4.6	10.7	14.9	16.6	19.6	17.9	11.5	4.0	-1.6
		MCWB	-3.1	-4.5	-1.9	1.8	7.2	11.9	14.7	16.1	13.9	8.2	2.1	-2.1
	10%	DB	-4.8	-5.8	-1.7	2.6	7.9	12.5	14.8	17.4	15.8	9.7	2.4	-3.2
		MCWB	-5.4	-6.7	-2.7	0.8	5.6	10.4	13.3	15.2	12.9	7.5	1.0	-3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	-1.8	0.8	6.9	12.1	16.6	18.7	20.1	17.3	12.0	5.7	0.7
		MCDB	-0.7	-1.2	2.6	11.4	17.8	23.7	22.7	24.3	21.0	16.3	6.8	0.8
	2%	WB	-1.9	-3.1	-0.7	3.9	9.4	14.0	16.5	18.0	15.8	10.4	4.2	-0.6
		MCDB	-1.7	-2.5	0.2	7.4	13.5	17.9	19.0	21.6	19.0	12.9	5.2	-0.4
	5%	WB	-3.1	-4.6	-1.6	2.1	7.4	12.2	14.9	16.6	14.4	9.1	2.4	-2.0
		MCDB	-2.6	-3.8	-0.9	4.2	10.4	14.6	16.5	19.1	17.3	10.8	3.5	-1.7
	10%	WB	-5.4	-6.6	-2.6	0.9	5.7	10.6	13.5	15.3	13.3	7.8	1.1	-3.9
		MCDB	-4.8	-5.9	-1.8	2.2	7.8	12.3	14.7	17.2	15.5	9.4	2.2	-3.3
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.9	10.1	6.9	6.0	6.2	5.5	6.5	8.5	8.9	8.3	8.9
		MCDBR	6.5	8.6	10.3	11.9	12.9	12.4	10.2	11.7	12.8	12.8	9.0	6.4
		MCWBR	6.5	8.0	8.6	7.8	8.0	7.9	7.2	7.4	7.3	8.6	7.1	6.0
	5% WB	MCDBR	6.4	7.8	8.7	10.7	12.3	11.4	9.6	10.7	10.8	11.2	7.6	5.4
		MCWBR	6.4	7.4	7.5	7.3	8.1	7.6	7.1	7.3	6.9	8.2	6.5	5.5
Clear-Sky Solar Irradiance	taub	0.285	0.313	0.379	0.471	0.502	0.466	0.448	0.422	0.377	0.385	0.348	0.298	
	taud	2.061	1.984	1.825	1.693	1.851	2.134	2.253	2.343	2.407	2.190	2.123	2.088	
	Ebn at Noon	713	790	793	759	761	798	808	812	814	707	625	629	
	Edh at Noon	84	118	169	217	194	147	129	113	95	98	80	71	
All-Sky Solar Radiation	RadAvg	1.31	2.43	3.69	4.46	4.84	5.27	4.76	4.15	3.43	2.06	1.11	0.89	
	RadStd	0.13	0.19	0.27	0.45	0.49	0.52	0.59	0.35	0.28	0.18	0.12	0.11	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (4 neighbors)	+0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page